



Simultaneous state and parameter estimation

- You have now learned how to use Kalman filters to perform state estimation and parameter estimation independently.
 - How about both at the same time?
- There are two approaches to doing so: joint estimation and dual estimation.
- I find that joint estimation (usually) works better and is easier to understand, so that is what I will focus on this week.



Generic joint estimation

- In joint estimation, state and parameter vectors are combined, KF simultaneously estimates this augmented state vector:

$$\begin{bmatrix} x_k \\ \theta_k \end{bmatrix} = \begin{bmatrix} f(x_{k-1}, u_{k-1}, \theta_{k-1}, w_{k-1}) \\ \theta_{k-1} + r_{k-1} \end{bmatrix}$$

$$z_k = h(x_k, u_k, \theta_k, v_k).$$

- To simplify notation, let $\mathbb{X}_k$ be the augmented state, $\mathbb{W}_k$ be the augmented noise, and $\mathcal{F}$ be the augmented state equations:

$$\mathbb{X}_k = \mathcal{F}(\mathbb{X}_{k-1}, u_{k-1}, \mathbb{W}_{k-1})$$

$$z_k = h(\mathbb{X}_k, u_k, v_k).$$



Evaluating generic joint estimation

- Advantage:
 - Quite straightforward to implement: with augmented model of system state dynamics and parameter dynamics defined, we simply apply a nonlinear KF method.
- Disadvantages:
 - Large matrix operations due to the high dimensionality of the resulting augmented model.
 - Potentially poor numeric conditioning due to the vastly different time scales of the states/parameters in augmented state vector.



Generic dual estimation: state evolution

- In dual estimation, separate KFs are used for state and parameter estimation.
- Computational complexity lower, matrices may be numerically better conditioned.
- However, by decoupling state from parameters, any cross-correlations between changes are lost, leading to potentially poorer accuracy.
- Model of state dynamics again explicitly includes parameters as vector θ_k :

$$x_k = f(x_{k-1}, u_{k-1}, w_{k-1}, \theta_{k-1})$$

$$y_k = h(x_k, u_k, v_k, \theta_{k-1}).$$

- Non-time-varying numeric values required by the model may be embedded within $f(\cdot)$ and $h(\cdot)$, and are not included in θ_k .



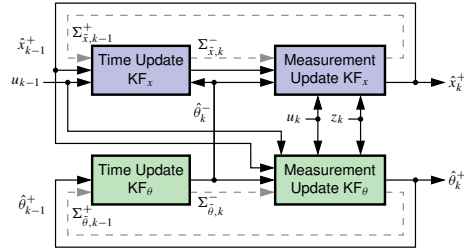
Generic dual estimation: parameter evolution

- Also slightly revise model of parameter dynamics to include effect of state equation explicitly:

$$\theta_k = \theta_{k-1} + r_{k-1}$$

$$d_k = h(f(x_{k-1}, u_{k-1}, \bar{w}_{k-1}, \theta_{k-1}), u_k, e_k, \theta_{k-1})$$

- The dual filters can be viewed by drawing a block diagram.
 - The interactions will be made clearer later.
- The process essentially comprises two KFs running in parallel—one adapting state and one adapting parameters—with some information exchange.



Dual estimation using EKF

- When finding $\hat{C}_k^\theta$, total-differential expansion must be correct:

$$\hat{C}_k^\theta = \left. \frac{dh(\hat{x}_k^-, u_k, \theta)}{d\theta} \right|_{\theta=\hat{\theta}_k^-}$$

$$\frac{dh(\hat{x}_k^-, u_k, \theta)}{d\theta} = \frac{\partial h(\hat{x}_k^-, u_k, \theta)}{\partial \theta} + \frac{\partial h(\hat{x}_k^-, u_k, \theta)}{\partial \hat{x}_k^-} \frac{d\hat{x}_k^-}{d\theta}$$

$$\frac{d\hat{x}_k^-}{d\theta} = \frac{\partial f(\hat{x}_{k-1}^+, u_{k-1}, \theta)}{\partial \theta} + \frac{\partial f(\hat{x}_{k-1}^+, u_{k-1}, \theta)}{\partial \hat{x}_{k-1}^+} \frac{d\hat{x}_{k-1}^+}{d\theta}$$

$$\frac{d\hat{x}_{k-1}^+}{d\theta} = \frac{d\hat{x}_{k-1}^-}{d\theta} - L_{k-1}^x \frac{dh(\hat{x}_{k-1}^-, u_{k-1}, \theta)}{d\theta}$$

- Assumes L_{k-1}^x not a function of θ (it is—weakly—not worth extra computation).
- The three total derivatives are computed recursively, initialized to zero.



Summary

- Two approaches to adapting a model's states and parameters at same time.
- Joint estimation augments both in large state vector, uses single standard nonlinear KF with this augmented system.
- Dual estimation uses separate nonlinear KFs for state and parameter estimation.
- Joint estimation easier conceptually but can be slower due to large matrix operations and possibly poorer matrix conditioning.
- Dual estimation generally faster, but ignores any correlation between state and parameter estimation in its updates.



Joint EKF for state and parameter estimation (1)

State-space model:

$$\begin{bmatrix} x_k \\ \theta_k \end{bmatrix} = \begin{bmatrix} f(x_{k-1}, u_{k-1}, w_{k-1}, \theta_{k-1}) \\ \theta_{k-1} + r_{k-1} \end{bmatrix} \quad \text{or} \quad \begin{aligned} \mathbb{X}_k &= \mathcal{F}(\mathbb{X}_{k-1}, u_{k-1}, \mathbb{W}_{k-1}) \\ d_k &= h(\mathbb{X}_k, u_k, v_k), \end{aligned}$$

where w_k , r_k , and v_k are independent, Gaussian noise processes of covariance matrices $\Sigma_{\tilde{w}}$, $\Sigma_{\tilde{r}}$, and $\Sigma_{\tilde{v}}$, respectively

For brevity, we let $\mathbb{X}_k = [x_k^T, \theta_k^T]^T$, $\mathbb{W}_k = [w_k^T, r_k^T]^T$ and $\Sigma_{\tilde{\mathbb{W}}} = \text{diag}(\Sigma_{\tilde{w}}, \Sigma_{\tilde{r}})$

Definitions:

$$\begin{aligned} \hat{A}_k &= \left. \frac{d\mathcal{F}(\mathbb{X}_k, u_k, \mathbb{W}_k)}{d\mathbb{X}_k} \right|_{\mathbb{X}_k = \hat{\mathbb{X}}_k^+} & \hat{B}_k &= \left. \frac{d\mathcal{F}(\mathbb{X}_k, u_k, \mathbb{W}_k)}{d\mathbb{W}_k} \right|_{\mathbb{W}_k = \tilde{\mathbb{W}}_k} \\ \hat{C}_k &= \left. \frac{dh(\mathbb{X}_k, u_k, v_k)}{d\mathbb{X}_k} \right|_{\mathbb{X}_k = \hat{\mathbb{X}}_k^-} & \hat{D}_k &= \left. \frac{dh(\mathbb{X}_k, u_k, v_k)}{dv_k} \right|_{v_k = \tilde{v}_k} \end{aligned}$$



Joint EKF for state and parameter estimation (2)

Initialization: For $k = 0$, set

$$\begin{aligned} \hat{\mathbb{X}}_0^+ &= \mathbb{E}[\mathbb{X}_0] \\ \Sigma_{\mathbb{X},0}^+ &= \mathbb{E}[(\mathbb{X}_0 - \hat{\mathbb{X}}_0^+)(\mathbb{X}_0 - \hat{\mathbb{X}}_0^+)^T] \end{aligned}$$

Computation: For $k = 1, 2, \dots$ compute:

$$\begin{aligned} \text{State estimate time update:} & \quad \hat{\mathbb{X}}_k^- = \mathcal{F}(\hat{\mathbb{X}}_{k-1}^+, u_{k-1}, \tilde{\mathbb{W}}_{k-1}) \\ \text{Error covariance time update:} & \quad \Sigma_{\mathbb{X},k}^- = \hat{A}_{k-1} \Sigma_{\mathbb{X},k-1}^+ \hat{A}_{k-1}^T + \hat{B}_{k-1} \Sigma_{\tilde{\mathbb{W}}} \hat{B}_{k-1}^T \\ \text{Output estimate:} & \quad \hat{d}_k = h(\hat{\mathbb{X}}_k^-, u_k, \tilde{v}_k) \\ \text{Estimator gain matrix:} & \quad L_k = \Sigma_{\mathbb{X},k}^- \hat{C}_k^T [\hat{C}_k \Sigma_{\mathbb{X},k}^- \hat{C}_k^T + \hat{D}_k \Sigma_{\tilde{v}} \hat{D}_k^T]^{-1} \\ \text{State estimate meas. update:} & \quad \hat{\mathbb{X}}_k^+ = \hat{\mathbb{X}}_k^- + L_k (d_k - \hat{d}_k) \\ \text{Error covariance meas. update:} & \quad \Sigma_{\mathbb{X},k}^+ = \Sigma_{\mathbb{X},k}^- - L_k \Sigma_{\tilde{d},k} L_k^T \end{aligned}$$



Joint SPKF for state and parameter estimation (1)

State-space model:

$$\begin{bmatrix} x_k \\ \theta_k \end{bmatrix} = \begin{bmatrix} f(x_{k-1}, u_{k-1}, w_{k-1}, \theta_{k-1}) \\ \theta_{k-1} + r_{k-1} \end{bmatrix} \quad \text{or} \quad \begin{aligned} \mathbb{X}_k &= \mathcal{F}(\mathbb{X}_{k-1}, u_{k-1}, \mathbb{W}_{k-1}) \\ z_k &= h(\mathbb{X}_k, u_k, v_k), \end{aligned}$$

$$z_k = h(x_k, u_k, v_k, \theta_k),$$

where w_k , r_k , and v_k are independent, Gaussian noise processes with means $\bar{w}$, $\bar{r}$, and $\bar{v}$, and covariance matrices $\Sigma_{\bar{w}}$, $\Sigma_{\bar{r}}$, and $\Sigma_{\bar{v}}$, respectively. For brevity, we let $\mathbb{X}_k = [x_k^T, \theta_k^T]^T$, $\mathbb{W}_k = [w_k^T, r_k^T]^T$ and $\Sigma_{\bar{\mathbb{W}}} = \text{diag}(\Sigma_{\bar{w}}, \Sigma_{\bar{r}})$

Definitions:

$$\mathbb{X}_k^a = [\mathbb{X}_k^T, \mathbb{W}_k^T, v_k^T]^T, \quad \mathcal{X}_k^a = [(\mathcal{X}_k^{\mathbb{X}})^T, (\mathcal{X}_k^{\mathbb{W}})^T, (\mathcal{X}_k^v)^T]^T, \quad p = 2 \times \dim(\mathbb{X}_k^a)$$



Joint SPKF for state and parameter estimation (2)

Initialization: For $k = 0$, set

$$\begin{aligned} \hat{\mathbb{X}}_0^+ &= \mathbb{E}[\mathbb{X}_0] \\ \Sigma_{\mathbb{X},0}^+ &= \mathbb{E}[(\mathbb{X}_0 - \hat{\mathbb{X}}_0^+)(\mathbb{X}_0 - \hat{\mathbb{X}}_0^+)^T] \\ \hat{\mathbb{X}}_0^{a,+} &= \mathbb{E}[\mathbb{X}_0^a] = [(\hat{\mathbb{X}}_0^+)^T, \bar{\mathbb{W}}, \bar{v}]^T \\ \Sigma_{\mathbb{X},0}^{a,+} &= \mathbb{E}[(\mathbb{X}_0^a - \hat{\mathbb{X}}_0^{a,+})(\mathbb{X}_0^a - \hat{\mathbb{X}}_0^{a,+})^T] \\ &= \text{diag}(\Sigma_{\mathbb{X},0}^+, \Sigma_{\bar{\mathbb{W}}}, \Sigma_{\bar{v}}) \end{aligned}$$

Computation: For $k = 1, 2, \dots$ compute:

$$\begin{aligned} \text{State time update:} \quad \mathcal{X}_{k-1}^{a,+} &= \left\{ \hat{\mathbb{X}}_{k-1}^{a,+}, \hat{\mathbb{X}}_{k-1}^{a,+} + \gamma \sqrt{\Sigma_{\mathbb{X},k-1}^{a,+}}, \hat{\mathbb{X}}_{k-1}^{a,+} - \gamma \sqrt{\Sigma_{\mathbb{X},k-1}^{a,+}} \right\} \\ \mathcal{X}_{k,i}^{\mathbb{X},-} &= \mathcal{F}(\mathcal{X}_{k-1,i}^{\mathbb{X},+}, u_{k-1}, \mathcal{X}_{k-1,i}^{\mathbb{W},+}) \\ \hat{\mathbb{X}}_k^- &= \sum_{i=0}^p \alpha_i^{(m)} \mathcal{X}_{k,i}^{\mathbb{X},-} \\ \text{Covariance time update:} \quad \Sigma_{\mathbb{X},k}^- &= \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{X}_{k,i}^{\mathbb{X},-} - \hat{\mathbb{X}}_k^-)(\mathcal{X}_{k,i}^{\mathbb{X},-} - \hat{\mathbb{X}}_k^-)^T \end{aligned}$$



Joint SPKF for state and parameter estimation (3)

Computation (cont.): For $k = 1, 2, \dots$ compute:

$$\begin{aligned} \text{Output estimate:} \quad \mathcal{Z}_{k,i} &= h(\mathcal{X}_{k,i}^{\mathbb{X},-}, u_k, \mathcal{X}_{k-1,i}^{v,+}) \\ \hat{z}_k &= \sum_{i=0}^p \alpha_i^{(m)} \mathcal{Z}_{k,i} \\ \text{Estimator gain matrix:} \quad \Sigma_{\tilde{z},k} &= \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{Z}_{k,i} - \hat{z}_k)(\mathcal{Z}_{k,i} - \hat{z}_k)^T \\ \Sigma_{\mathbb{X}\tilde{z},k}^- &= \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{X}_{k,i}^{\mathbb{X},-} - \hat{\mathbb{X}}_k^-)(\mathcal{Z}_{k,i} - \hat{z}_k)^T \\ L_k &= \Sigma_{\mathbb{X}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1} \\ \text{State estimate meas. update:} \quad \hat{\mathbb{X}}_k^+ &= \hat{\mathbb{X}}_k^- + L_k (z_k - \hat{z}_k) \\ \text{Error covariance meas. update:} \quad \Sigma_{\mathbb{X},k}^+ &= \Sigma_{\mathbb{X},k}^- - L_k \Sigma_{\tilde{z},k} L_k^T \end{aligned}$$



Dual EKF for state and parameter estimation (1)

State-space model:

$$\begin{aligned} x_{k+1} &= f(x_k, u_k, \theta_k, w_k) & \text{and} & & \theta_{k+1} &= \theta_k + r_k, \\ y_k &= h(x_k, u_k, \theta_k, v_k) & & & d_k &= h(x_k, u_k, \theta_k, e_k) \end{aligned}$$

where w_k , v_k , r_k and e_k are independent Gaussian noise processes with means $\bar{w}$, $\bar{v}$, $\bar{r}$, and $\bar{e}$ and covariance matrices $\Sigma_{\bar{w}}$, $\Sigma_{\bar{v}}$, $\Sigma_{\bar{r}}$ and $\Sigma_{\bar{e}}$, respectively

Definitions:

$$\begin{aligned} \hat{A}_k &= \left. \frac{df(x_k, u_k, \hat{\theta}_k^-, w_k)}{dx_k} \right|_{x_k = \hat{x}_k^+} & \hat{B}_k &= \left. \frac{df(x_k, u_k, \hat{\theta}_k^-, w_k)}{dw_k} \right|_{w_k = \bar{w}} \\ \hat{C}_k^x &= \left. \frac{dh(x_k, u_k, \hat{\theta}_k^-, v_k)}{dx_k} \right|_{x_k = \hat{x}_k^-} & \hat{D}_k^x &= \left. \frac{dh(x_k, u_k, \hat{\theta}_k^-, v_k)}{dv_k} \right|_{v_k = \bar{v}} \\ \hat{C}_k^\theta &= \left. \frac{dh(\hat{x}_k^-, u_k, \theta, e_k)}{d\theta} \right|_{\theta = \hat{\theta}_k^-} & \hat{D}_k^\theta &= \left. \frac{dh(\hat{x}_k^-, u_k, \theta, e_k)}{de_k} \right|_{e_k = \bar{e}} \end{aligned}$$



Dual EKF for state and parameter estimation (2)

Initialization: For $k = 0$, set

$$\begin{aligned} \hat{\theta}_0^+ &= \mathbb{E}[\theta_0], & \Sigma_{\bar{\theta},0}^+ &= \mathbb{E}[(\theta_0 - \hat{\theta}_0^+)(\theta_0 - \hat{\theta}_0^+)^T], \\ \hat{x}_0^+ &= \mathbb{E}[x_0], & \Sigma_{\bar{x},0}^+ &= \mathbb{E}[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T]. \end{aligned}$$

Computation: For $k = 1, 2, \dots$ compute:

$$\begin{aligned} \text{Time update for weight filter:} & \quad \hat{\theta}_k^- = \hat{\theta}_{k-1}^+ \\ & \quad \Sigma_{\bar{\theta},k}^- = \Sigma_{\bar{\theta},k-1}^+ + \Sigma_{\bar{r}} \\ \text{Time update for state filter:} & \quad \hat{x}_k^- = f(\hat{x}_{k-1}^+, u_{k-1}, \hat{\theta}_k^-, \bar{w}) \\ & \quad \Sigma_{\bar{x},k}^- = \hat{A}_{k-1} \Sigma_{\bar{x},k-1}^+ \hat{A}_{k-1}^T + \hat{B}_{k-1} \Sigma_{\bar{w}} \hat{B}_{k-1}^T \\ \text{Meas. update for state filter:} & \quad L_k^x = \Sigma_{\bar{x},k}^- (\hat{C}_k^x)^T [\hat{C}_k^x \Sigma_{\bar{x},k}^- (\hat{C}_k^x)^T + \hat{D}_k^x \Sigma_{\bar{v}} (\hat{D}_k^x)^T]^{-1} \\ & \quad \hat{x}_k^+ = \hat{x}_k^- + L_k^x [z_k - h(\hat{x}_k^-, u_k, \hat{\theta}_k^-, \bar{v})] \\ & \quad \Sigma_{\bar{x},k}^+ = \Sigma_{\bar{x},k}^- - L_k^x \Sigma_{\bar{z},k} (L_k^x)^T \end{aligned}$$



Dual EKF for state and parameter estimation (3)

Computation (cont.): For $k = 1, 2, \dots$ compute:

$$\begin{aligned} \text{Meas. update for weight filter:} & \quad L_k^\theta = \Sigma_{\bar{\theta},k}^- (\hat{C}_k^\theta)^T [\hat{C}_k^\theta \Sigma_{\bar{\theta},k}^- (\hat{C}_k^\theta)^T + \hat{D}_k^\theta \Sigma_{\bar{e}} (\hat{D}_k^\theta)^T]^{-1} \\ & \quad \hat{\theta}_k^+ = \hat{\theta}_k^- + L_k^\theta [z_k - h(\hat{x}_k^-, u_k, \hat{\theta}_k^-, \bar{e})] \\ & \quad \Sigma_{\bar{\theta},k}^+ = \Sigma_{\bar{\theta},k}^- - L_k^\theta \Sigma_{\bar{z},k} (L_k^\theta)^T \end{aligned}$$



Dual SPKF for state and parameter estimation (1)

State-space model:

$$\begin{aligned} x_k &= f(x_{k-1}, u_{k-1}, w_{k-1}, \theta_{k-1}) & \text{and} & & \theta_k &= \theta_{k-1} + r_{k-1}, \\ z_k &= h(x_k, u_k, v_k, \theta_k) & & & d_k &= h(x_k, u_k, \bar{v}_k, \theta_{k-1}, e_k) \end{aligned}$$

where w_k , v_k , r_k and e_k are independent, Gaussian noise processes with means $\bar{w}$, $\bar{v}$, $\bar{r}$, and $\bar{e}$, and covariance matrices $\Sigma_{\bar{w}}$, $\Sigma_{\bar{v}}$, $\Sigma_{\bar{r}}$ and $\Sigma_{\bar{e}}$, respectively

Definitions:

$$x_k^a = [x_k^T, w_k^T, v_k^T]^T, \quad \mathcal{X}_k^a = [(\mathcal{X}_k^x)^T, (\mathcal{X}_k^w)^T, (\mathcal{X}_k^v)^T]^T, \quad p = 2 \times \dim(x_k^a)$$



Dual SPKF for state and parameter estimation (2)

Initialization: For $k = 0$, set

$$\begin{aligned} \hat{\theta}_0^+ &= \mathbb{E}[\theta_0] \\ \hat{x}_0^+ &= \mathbb{E}[x_0] \\ \Sigma_{\tilde{x},0}^+ &= \mathbb{E}[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T] \\ \Sigma_{\tilde{\theta},0}^+ &= \mathbb{E}[(\theta_0 - \hat{\theta}_0^+)(\theta_0 - \hat{\theta}_0^+)^T] \end{aligned} \quad \begin{aligned} \hat{x}_0^{a,+} &= \mathbb{E}[x_0^a] = [(\hat{x}_0^+)^T, \bar{w}, \bar{v}]^T \\ \Sigma_{\tilde{x},0}^{a,+} &= \mathbb{E}[(x_0^a - \hat{x}_0^{a,+})(x_0^a - \hat{x}_0^{a,+})^T] \\ &= \text{diag}(\Sigma_{\tilde{x},0}^+, \Sigma_w, \Sigma_v) \end{aligned}$$

Computation: For $k = 1, 2, \dots$ compute:

$$\begin{aligned} \text{Parameter estimate time update:} \quad \hat{\theta}_k^- &= \hat{\theta}_{k-1}^+ \\ \text{Parameter covariance time update:} \quad \Sigma_{\tilde{\theta},k}^- &= \Sigma_{\tilde{\theta},k-1}^+ + \Sigma_{\tilde{r}} \end{aligned}$$



Dual SPKF for state and parameter estimation (3)

Computation (cont.): For $k = 1, 2, \dots$ compute:

$$\begin{aligned} \text{State time update:} \quad \mathcal{X}_{k-1}^{a,+} &= \left\{ \hat{x}_{k-1}^{a,+}, \hat{x}_{k-1}^{a,+} + \gamma \sqrt{\Sigma_{\tilde{x},k-1}^{a,+}}, \hat{x}_{k-1}^{a,+} - \gamma \sqrt{\Sigma_{\tilde{x},k-1}^{a,+}} \right\} \\ \mathcal{X}_{k,i}^{x,-} &= f(\mathcal{X}_{k-1,i}^{x,+}, u_{k-1}, \mathcal{X}_{k-1,i}^{w,+}, \hat{\theta}_k^-) \\ \hat{x}_k^- &= \sum_{i=0}^p \alpha_i^{(m)} \mathcal{X}_{k,i}^{x,-} \\ \text{Covariance time update:} \quad \Sigma_{\tilde{x},k}^- &= \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{X}_{k,i}^{x,-} - \hat{x}_k^-)(\mathcal{X}_{k,i}^{x,-} - \hat{x}_k^-)^T \\ \text{Output, parameter filter:} \quad \mathcal{W}_k &= \left\{ \hat{\theta}_k^-, \hat{\theta}_k^- + \gamma \sqrt{\Sigma_{\tilde{\theta},k}^-}, \hat{\theta}_k^- - \gamma \sqrt{\Sigma_{\tilde{\theta},k}^-} \right\} \\ \mathcal{D}_{k,i} &= h(f(\hat{x}_{k-1}^+, u_{k-1}, \bar{w}_{k-1}, \mathcal{W}_{k,i}), u_k, \bar{v}_k, \mathcal{W}_{k,i}) \\ \hat{d}_k &= \sum_{i=0}^p \alpha_i^{(m)} \mathcal{D}_{k,i} \\ \text{Output, state filter:} \quad \mathcal{Z}_{k,i} &= h(\mathcal{X}_{k,i}^{x,-}, u_k, \mathcal{X}_{k-1,i}^{v,+}, \hat{\theta}_k^-) \\ \hat{z}_k &= \sum_{i=0}^p \alpha_i^{(m)} \mathcal{Z}_{k,i} \end{aligned}$$



Dual SPKF for state and parameter estimation (4)

Computation (cont.): For $k = 1, 2, \dots$ compute:

$$\begin{aligned}
 \text{State filter gain matrix: } \quad & \Sigma_{\tilde{z},k} = \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{Z}_{k,i} - \hat{z}_k) (\mathcal{Z}_{k,i} - \hat{z}_k)^T \\
 & \Sigma_{\tilde{x}\tilde{z},k}^- = \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{X}_{k,i}^{x,-} - \hat{x}_k^-) (\mathcal{Z}_{k,i} - \hat{z}_k)^T \\
 & L_k^x = \Sigma_{\tilde{x}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1} \\
 \text{Param. filter gain matrix: } \quad & \Sigma_{\tilde{d},k} = \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{D}_{k,i} - \hat{d}_k) (\mathcal{D}_{k,i} - \hat{d}_k)^T \\
 & \Sigma_{\tilde{\theta}\tilde{d},k}^- = \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{W}_{k,i} - \hat{\theta}_k^-) (\mathcal{D}_{k,i} - \hat{d}_k)^T \\
 & L_k^\theta = \Sigma_{\tilde{\theta}\tilde{d},k}^- \Sigma_{\tilde{d},k}^{-1} \\
 \text{State meas. update: } \quad & \hat{x}_k^+ = \hat{x}_k^- + L_k^x (\mathcal{z}_k - \hat{z}_k) \\
 \text{Covariance meas. update: } \quad & \Sigma_{\tilde{x},k}^+ = \Sigma_{\tilde{x},k}^- - L_k^x \Sigma_{\tilde{z},k} (L_k^x)^T \\
 \text{Parameter meas. update: } \quad & \hat{\theta}_k^+ = \hat{\theta}_k^- + L_k^\theta (\mathcal{z}_k - \hat{d}_k) \\
 \text{Covariance meas. update: } \quad & \Sigma_{\tilde{\theta},k}^+ = \Sigma_{\tilde{\theta},k}^- - L_k^\theta \Sigma_{\tilde{d},k} (L_k^\theta)^T
 \end{aligned}$$